

Roll No.

BEM–C102/BEM–C101(O)

B. TECH. (FIRST SEMESTER) (REPEAT EXAM.)

EXAMINATION, 2021

SUBJECT : ENGINEERING MATHEMATICS–I

Time : 3 Hours

Maximum Marks : 70

Maximum Pass Marks : 40%

Note : Question paper is divided into two Sections : A and B. Attempt both the Sections as per given instructions.

Section—A

(Short Answer Type Questions)

Note : Answer any *five* questions in about 150 words each. Each question carries 6 marks. (5 × 6 = 30 Marks)

1. Find the n th derivative of $\frac{x}{x^2 + a^2}$.

2. Find the radius of curvature to the cycloid :

$$x = a (\theta + \sin \theta)$$

and

$$y = a (1 - \cos \theta)$$

at the point θ . What will it be at $\theta = \pi$?

3. If $u = \tan^{-1} \left(\frac{x^2 + y^2}{x - y} \right)$, prove that :

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u.$$

4. Verify whether :

$$u = \sin^{-1} x + \sin^{-1} y$$

and

$$v = x\sqrt{1 - y^2} + y\sqrt{1 - x^2}$$

are functionally dependent, and if so find the relation between them.

5. Evaluate by changing the order of integration $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$.

6. Evaluate :

$$\int_0^1 x^{n-1} \left[\log \left(\frac{1}{x} \right) \right]^{m-1} dx.$$

7. Use Green's theorem to evaluate :

$$\int_C [(x^2 + xy) dx + (x^2 + y^2) dy]$$

where C is a square formed by the lines $x = \pm 1, y = \pm 1$.

8. State and prove physical significance of CURL.
 9. Show that the matrix $B^0 AB$ is Hermitian or skew-Hermitian according as A is Hermitian or skew-Hermitian.
 10. Reduce the matrix A into its normal form :

$$A = \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

Section—B**(Long Answer Type Questions)**

Note : Answer any *four* questions in detail. Each question carries 10 marks.

(4 × 10 = 40 Marks)

11. If $\cos^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n$, prove that :

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + 2n^2 y_n = 0.$$

12. A function $f(x, y)$ is rewritten in terms of new variables $u = e^x \cos y$, $v = e^x \sin y$, prove that :

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = (u^2 + v^2) \left(\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} \right).$$

13. Use the method of Lagrange's multipliers to find volume of largest rectangular parallelopiped that can be inscribed in the ellipsoid :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

14. Evaluate :

$$\iiint_S xyz \, dx \, dy \, dz,$$

where $S = [(x, y, z): x^2 + y^2 + z^2 \leq 1, \, x \geq 0, \, y \geq 0, \, z \geq 0]$.

15. A pyramid is bounded by the three coordinate planes and the plane $x + 2y + 3z = 6$. Compute this volume by double integration.

16. Verify Stokes' theorem for $\vec{F} = (x + y)\mathbf{i} + (2x - z)\mathbf{j} + (y + z)\mathbf{k}$ for the surface of a triangular lamina with vertices $(2, 0, 0)$, $(0, 3, 0)$, $(0, 0, 6)$.

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17. Verify Cayley-Hamilton theorem for the matrix :

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

Express $A^6 - 6A^5 + 9A^4 - 2A^3 - 12A^2 + 23A - 9I$ as linear polynomial in A.

18. Test for consistency and solve :

$$5x + 3y + 7z = 4$$

$$3x + 26y + 2z = 9$$

$$7x + 2y + 10z = 5.$$

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